Ling 285
Human Language and Technology
Lecture 12

Masses and Springs in Speech Production

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ECONOMY

2 Americans win Nobel Prize in economics for work on boom-bust cycles

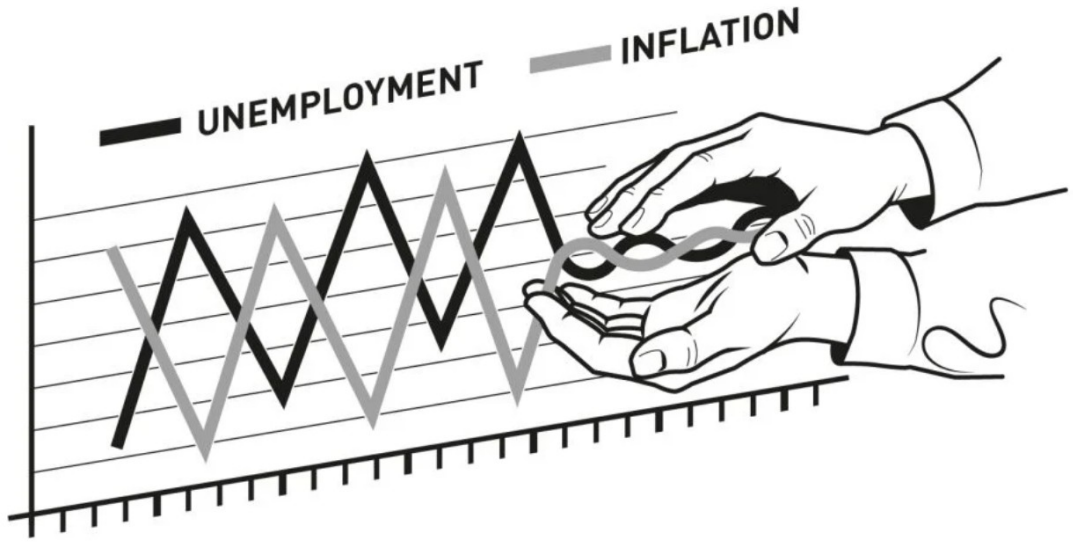
UPDATED OCTOBER 12, 2026 · 8:10 AM ET

HEARD ON MORNING EDITION

Scott Horsley

3-Minute Listen + PLAYLIST TRANSCRIPT ···





The aim of stabilisation policy is to even out peaks and troughs, to avoid problems such as high unemployment or runaway inflation. © Johan Jarnestad/The Royal Swedish Academy of Sciences

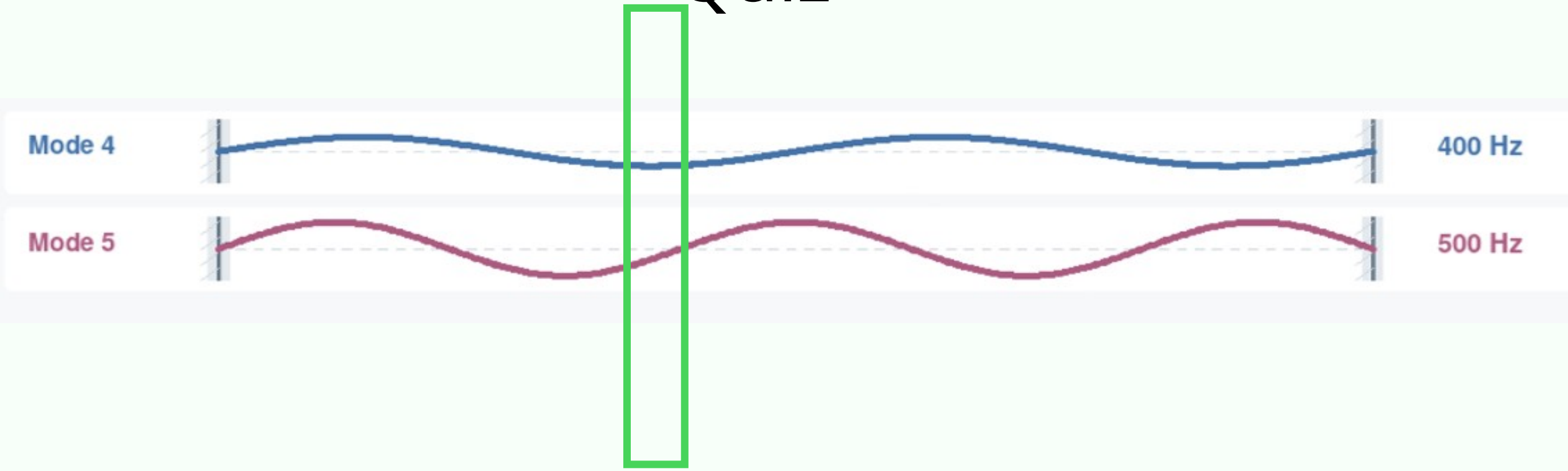
Woodford's Theory:
Dynamic Stochastic General Equilibrium (DSGE)

- https://www.nobelprize.org/prizes/economic-sciences/2026/popular-information/

Principles of Mass-Spring Systems

1. Mass ↑ Frequency ↓, Stiffness ↑ Frequency ↓
2. An n-mass system has n modes of vibration, each with a higher frequency
3. An m's increase lowers frequency proportionally to how much the mass moves, which depends on where it is!
4. Where there's maximal movement, there's no stretching, An where there's maximal stretching, there's no movement!
5. A k's increase raises frequency proportionally to how much the spring stretches/compresses, which depends on where it is!
6. In a Fixed-Fixed System, the harmonics are multiples of the base frequency, because the base mode fits n-times into each successive mode
7. In a Fixed-Free System, the harmonics are odd multiples of the base frequency, because the base mode fits an odd n-times into each successive mode

Pop Quiz



Increasing the Mass lowers mode 4's frequency, but doesn't affect mode 5's (why? Because Increaing mass
lowers freq If the mass is moving!). Increasing Stiffness doesn't affect mode 4, but increases mode 5. So an
increase in k and m there lowers Mode 4's frequency and raises Mode 5's frequency!

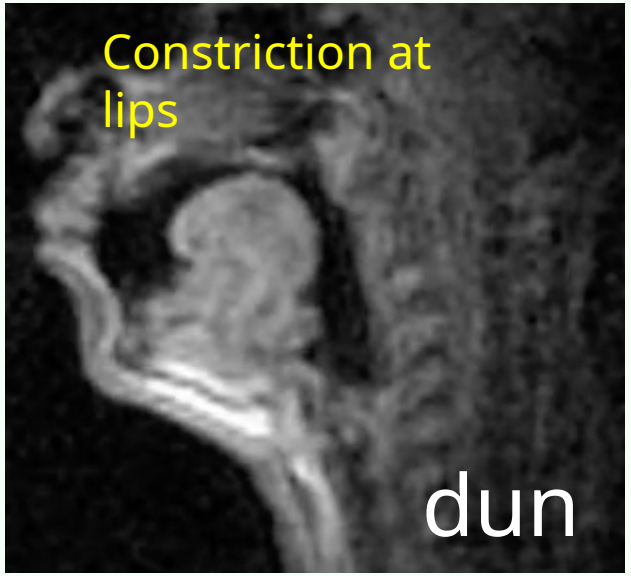
Theory, Prediction, and Experiment

- The Scientific Method depends on the statement of very clear theories about how the world is like, usually in the form of differential equations.
- We then solve these differential equations, and the result is a *prediction* of what the world is like when we actually measure it!
- Theory: The vocal folds are a membrane, so when it vibrates, the wave of vibration will have equidistant peaks in the spectrum (which we call harmonics).
- The differential equations are many of the type we have already seen, but slightly more complicated
- Prediction: if I measure any of your laryngeal vibration signals, they will have a spectrum with equidistant harmonics.
- Let’s do an experiment to find out if this is true!!!!!

Vowels

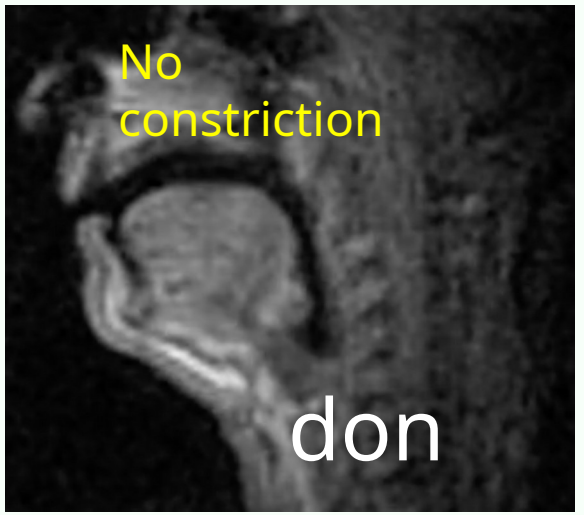
High





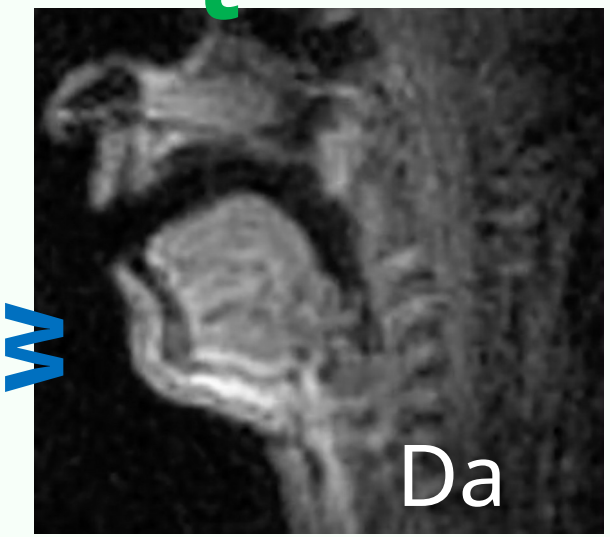
Front

Back



Central

Low





/u/ = free-end loading

Mode 1: motion dominates

Mode 2: motion dominates

F1 ↓ F2 ↓

Click to reveal the reasoning steps.

/i/ = constriction near 2/3

Mode 1: mass motion dominates

Mode 2: springiness dominates

F1 ↓ F2 ↑

Click to reveal the reasoning steps.

/a/ = constriction near 1/3

Mode 1: springiness dominates

Mode 2: mass motion dominates

F1 ↑ F2 ↓

Click to reveal the reasoning steps.

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Where were you a year ago?